

# FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

## Student Assessment Document

### FINAL EXPLANATION: ESSENTIAL MATHEMATICS – GRADE 10 | SUB-STRAND 2.5: AREA OF PART OF A CIRCLE

#### Student Assessment Document

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| Student Name | _____ |
| Class        | _____ |
| Date         | _____ |

#### INSTRUCTIONS FOR STUDENTS

This is your Final Explanation document. Your goal is to write a complete, well-organised answer to the driving question using everything you have learned across all 8 lessons. Read the driving question carefully, then use each section below as a building block for your full explanation. Write in full sentences and paragraphs. Show all working where calculations are required. Use diagrams where they help your explanation. A model exemplar answer is provided in each section — study it, then write your own response in your own words and using your own examples.

#### Section 1: Anchor Your Explanation to the Phenomenon

**Start by explaining the real-world puzzle that launched this unit. Describe what is happening at Wakulima Market and why the mama mboga's watermelon slices look so different in size even when they all come from the same circular melon. What is the key variable that controls how big each wedge is? Why can't we use the area formulas we already know — for rectangles, triangles, or other polygons — to solve this problem directly?**

At Wakulima Market in Nairobi, a mama mboga cuts a large circular watermelon into wedge-shaped slices to sell by the piece. Even though every slice comes from the same melon — meaning every slice shares the same radius — customers notice that some pieces look much bigger than others. The key variable is the angle of the cut. A wide cut creates a large wedge; a narrow cut creates a thin sliver. This is puzzling because our earlier area formulas (for rectangles, triangles, and polygons) only work for shapes bounded by straight sides. A watermelon wedge is bounded by two straight radii AND a curved arc (part of the circle's circumference), so none of our old formulas apply directly. We need a new approach that accounts for the curved boundary and the size of the central angle.

#### Section 2: Define the Key Terms and Shapes

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| Clearly define each of the | A sector is the 'wedge' shape formed by two radii and the arc between them — like one slice of the watermelon. A segment is the |
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| <p>following terms and support each definition with a labelled sketch or a real-life example from the Kenyan context: (a) Sector, (b) Segment, (c) Arc, (d) Central angle (in both degrees and radians), (e) Minor vs. major sector. Explain the difference between a sector and a segment — many students confuse these two shapes.</p> | <p>region between a chord (a straight cut that does NOT pass through the centre) and the arc it cuts off — like the rind-edge piece left over if you cut straight across rather than through the centre. An arc is the curved part of the circle's boundary between two points. The central angle is the angle at the centre of the circle formed by the two radii; it can be measured in degrees (e.g., 90°) or in radians (e.g., <math>\pi/2</math> rad). A minor sector has a central angle less than 180° (a small slice), while a major sector has a central angle greater than 180° (more than half the melon). The easiest way to remember the difference between a sector and a segment: a sector always includes the centre point of the circle; a segment never includes the centre point.</p> |
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| Section 3: Derive and Apply the Area Formulas                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| <p>Explain WHERE the formula for the area of a sector comes from — do not just state it. Show how it is derived from the area of a full circle. Then write the formula in BOTH degree form and radian form. Finally, show a fully worked example using realistic Kenyan numbers: a watermelon with radius 21 cm is cut into a sector with a central angle of 60°. Calculate the exact area of that slice. Then show how to find the area of a segment by combining the sector formula with the area of a triangle.</p> | <p>A full circle has area <math>A = \pi r^2</math>. A full circle also has a central angle of 360°. A sector is simply a fraction of the full circle — the fraction is <math>(\theta/360)</math> when <math>\theta</math> is in degrees. Therefore:</p> <p>Area of sector (degrees) = <math>(\theta/360) \times \pi r^2</math></p> <p>In radian form, because a full circle = <math>2\pi</math> radians, the fraction becomes <math>(\theta/2\pi)</math>, giving:</p> <p>Area of sector (radians) = <math>\frac{1}{2}r^2\theta</math></p> <p>Both formulas express the same idea: the sector's area is proportional to its central angle.</p> <p>Worked Example — Watermelon Slice:<br/> Radius <math>r = 21</math> cm, central angle <math>\theta = 60^\circ</math><br/> Area of sector = <math>(60/360) \times \pi \times 21^2</math><br/> <math>= (1/6) \times \pi \times 441</math><br/> <math>= 441\pi/6</math><br/> <math>= 73.5\pi</math><br/> <math>\approx 230.9 \text{ cm}^2</math></p> <p>So one 60° slice of the watermelon covers approximately 230.9 cm<sup>2</sup>.</p> <p>Area of a Segment:<br/> A segment = sector – triangle formed by the two radii and the chord.<br/> Area of triangle = <math>\frac{1}{2}r^2\sin \theta</math><br/> Area of segment = <math>\frac{1}{2}r^2\theta - \frac{1}{2}r^2\sin \theta = \frac{1}{2}r^2(\theta - \sin \theta)</math> [<math>\theta</math> in radians]</p> |

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|  | <p>For the same watermelon (<math>r = 21</math> cm, <math>\theta = 60^\circ = \pi/3</math> rad):</p> <p>Area of segment = <math>\frac{1}{2} \times 21^2 \times (\pi/3 - \sin 60^\circ)</math></p> <p>= <math>\frac{1}{2} \times 441 \times (1.047 - 0.866)</math></p> <p>= <math>220.5 \times 0.181</math></p> <p><math>\approx 39.9</math> cm<sup>2</sup></p> <p>This tells us how much watermelon flesh lies between the straight chord cut and the curved rind.</p> |
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#### Section 4: Answer the Driving Question Directly

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| <p><b>Now answer the driving question fully and directly. Explain how you can calculate the EXACT area of any sector or segment shape. Then explain how this knowledge solves the fair-sharing problem at Wakulima Market. If the mama mboga wants every customer to receive exactly the same area of watermelon, what must she control? Use mathematics to support your answer. Include at least one example showing that two slices with DIFFERENT angles from the SAME melon have DIFFERENT areas, proving that angle alone explains the size difference.</b></p> | <p>We can calculate the exact area of any wedge-shaped sector using the formula: Area = <math>(\theta/360) \times \pi r^2</math>, where <math>\theta</math> is the central angle in degrees and <math>r</math> is the radius of the circle. For a segment, we subtract the triangle: Area = <math>\frac{1}{2}r^2(\theta - \sin \theta)</math> with <math>\theta</math> in radians.</p> <p>This directly solves the Wakulima Market puzzle. The mama mboga's slices look different because she is cutting at different central angles. Since all slices come from the same melon (same <math>r</math>), the ONLY variable that changes the area is <math>\theta</math>.</p> <p>Proof with numbers (<math>r = 21</math> cm):</p> <ul style="list-style-type: none"> <li>– Slice A: <math>\theta = 40^\circ \rightarrow</math> Area = <math>(40/360) \times \pi \times 441 = 49\pi \approx 153.9</math> cm<sup>2</sup></li> <li>– Slice B: <math>\theta = 80^\circ \rightarrow</math> Area = <math>(80/360) \times \pi \times 441 = 98\pi \approx 307.9</math> cm<sup>2</sup></li> </ul> <p>Slice B is exactly TWICE the size of Slice A, even though both come from the same melon. Doubling the angle doubles the area — they are directly proportional.</p> <p>For a truly fair share: if the melon has a full angle of <math>360^\circ</math> and the mama mboga wants to give <math>n</math> customers equal slices, each slice must have a central angle of <math>(360^\circ \div n)</math>. For 8 customers: <math>360^\circ \div 8 = 45^\circ</math> per slice. Each slice would have area = <math>(45/360) \times \pi \times 21^2 = (1/8) \times 441\pi \approx 173.2</math> cm<sup>2</sup>. Mathematics gives the mama mboga — and every customer — a guarantee of fairness.</p> |
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#### Section 5: Extend to Design and Real-Life Applications

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| <p><b>Go beyond the market. Describe at least TWO other real-life situations in Kenya where calculating the area of a sector or segment is genuinely useful. For each situation, identify the shape (sector or segment?), state what <math>r</math> and <math>\theta</math> represent in that context, and explain what the calculated area tells us. Then reflect: What is</b></p> | <p>Application 1 — Irrigation Pivot Farming in the Rift Valley:</p> <p>In large-scale farms near Nakuru, centre-pivot irrigation systems rotate around a fixed water source and spray water in a circular arc. If a pivot arm of length 80 m rotates through an angle of <math>120^\circ</math>, the irrigated area is a sector: Area = <math>(120/360) \times \pi \times 80^2 = (1/3) \times \pi \times 6400 \approx 6702</math> m<sup>2</sup>. A farmer needs this calculation to know how many bags of fertiliser to apply and whether the water supply is sufficient for that exact area of crops.</p> <p>Application 2 — Designing a Kenyan Craft Fan or Kikoy Pattern:</p> <p>Artisans in Mombasa and Kisumu design decorative fans and woven kikoy borders using repeating sector and segment patterns. A designer creating a fan that opens to <math>150^\circ</math> with a radius of 30 cm needs the sector area to estimate fabric cost: Area = <math>(150/360) \times \pi \times 30^2 \approx 1178.1</math> cm<sup>2</sup>. Accurate costing depends on this calculation.</p> |
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| the most important mathematical idea you are taking away from this unit, and how has it changed the way you think about curved shapes in the world around you? | <p>Reflection:</p> <p>The most important idea from this unit is that area is proportional to angle when the radius stays fixed. Before this unit, curved boundaries felt impossible to measure exactly. Now I understand that a circle is just a shape whose parts can be unlocked by treating the central angle as a fraction of the whole <math>360^\circ</math>. Every arc, every wedge, every segment becomes calculable — and that makes the curved world around us just as measurable as the straight-edged world we started with.</p> |
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| RUBRIC                                     |                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                            |
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| Criterion                                  | Excellent (4)                                                                                                                                                                                                                                                                                                                                  | Proficient (3)                                                                                                                                                                                                                                                      | Developing (1–2)                                                                                                                                                                                                           |
| Phenomenon Connection                      | The student consistently and explicitly connects every section of their explanation back to the Wakulima Market watermelon phenomenon. The real-world puzzle is used as a thread that runs through the entire response, not just the introduction.                                                                                             | The student connects most sections to the watermelon phenomenon. The link is clear in the introduction and conclusion but may be absent from one or two middle sections.                                                                                            | The student mentions the phenomenon briefly in the introduction but then writes about mathematics in abstract terms without returning to the real-world context.                                                           |
| Conceptual Accuracy – Sectors and Segments | Sector and segment are defined precisely and correctly, with labelled diagrams. The student clearly distinguishes between the two shapes (e.g., sector includes the centre; segment does not) and correctly identifies which formula applies to which shape.                                                                                   | Sector and segment are defined correctly and the formulas are applied correctly, but the distinction between the two shapes may not be stated explicitly or diagrams may be missing or unlabelled.                                                                  | The student confuses sector and segment, or applies the segment formula where the sector formula is needed (or vice versa). Definitions are vague or missing.                                                              |
| Derivation and Formula Use                 | The student clearly derives the sector area formula from the full circle formula by reasoning about proportional fractions of $360^\circ$ (or $2\pi$ radians). Both degree and radian forms are stated. All worked calculations are correct, fully shown step by step, and include correct units ( $\text{cm}^2$ ).                            | The student states the formula correctly and shows most working steps. The derivation may be partially explained rather than fully developed. Minor arithmetic errors may appear but do not affect the overall logic.                                               | The student states the formula without derivation or explanation. Key steps are missing from working, or there are significant arithmetic errors. Units may be missing or incorrect.                                       |
| Answering the Driving Question             | The student answers the driving question directly, completely, and with mathematical evidence. They show with at least one calculation that changing the angle changes the area proportionally, and they explain exactly what the mama mboga must do to ensure fair sharing. The answer is logically structured and mathematically convincing. | The student answers the driving question and provides a calculation showing that angle affects area. The fair-sharing solution is identified but may not be fully developed mathematically (e.g., formula given but not applied to a specific number of customers). | The student attempts to answer the driving question but the response is vague, incomplete, or not supported by correct mathematical working. The connection between central angle and area may be stated without evidence. |
| Real-Life Extension and                    | The student identifies two or more                                                                                                                                                                                                                                                                                                             | The student identifies two Kenyan                                                                                                                                                                                                                                   | The student identifies only one                                                                                                                                                                                            |

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| Mathematical Reflection | distinct, realistic Kenyan applications (beyond the watermelon), correctly identifies $r$ and $\theta$ in each context, performs accurate area calculations, and writes a thoughtful reflection that articulates a transferable mathematical insight about proportionality and curved shapes. | applications and performs calculations, but one application may be weakly contextualised or the reflection may be superficial (e.g., 'I learned the formula for a sector'). | application or the applications given are not distinctly Kenyan or realistic. Calculations may be missing or incorrect. The reflection restates what was done rather than identifying a transferable mathematical idea. |
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## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

**Production & Layout:** Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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